Ant Colony Optimization for Wildfire Evacuation Routing over a Fire-Consuming Road Network

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Roll: 25

July 2026

Abstract

This report studies evacuation routing while a wildfire is burning through the road network. A fire starts near a wildland-urban-interface town and spreads outward, faster downwind, following the wind-stretched elliptical pattern used by operational fire simulators. Neighbourhood groups must drive to shelters, but each road segment has a fire arrival time and can be crossed only before the fire reaches it, allowing for a safety margin. The result is a time-dependent, NP-hard routing problem with no gradient and with discontinuous failure: one late segment can trap a group. We solve it with Ant Colony Optimization (ACO), implemented from scratch, checking feasibility against the fire clock at every construction step. We compare ACO against a fire-aware greedy router and a fire-blind shortest path under the same feasibility rules. In the demonstration scenario both baselines lose groups (18.96 and 11.72 penalized hours), while ACO evacuates all six groups in 5.02 hours, an improvement of 73.5% and 57.1%. Across 15 random scenarios ACO averages 3.54 hours against greedy's 12.18, with a Wilcoxon signed-rank $p \approx 0.0004$. Published ACO evacuation work is set indoors, in buildings, and published outdoor wildfire evacuation systems do not use ACO, so this combination addresses a gap confirmed by the literature review. The report gives the fire model, the ACO equations with every symbol defined, the constraint handling, the results, and the visualizations, including an animation of the fire front burning through the network while groups escape.

1 Problem Statement

A rural road network serves a town at the wildland-urban interface: a grid of intersections with a fraction of the segments missing, so the map has dead ends and forced detours like a real road system. A wildfire ignites upwind and spreads outward, and each intersection n has a fire arrival time $t_{\text{fire}}(n)$. Several neighbourhood groups must drive from their homes to designated shelters. A group may drive road (u, v) starting at time t only if it leaves u and reaches v before the fire arrives at each, minus a safety margin. The task is to route every group to a shelter and minimize total evacuation time, with a large penalty for any group that cannot reach safety.

1.1 Why this problem is real

Wildfire evacuation is a current operational problem. The January 2025 Palisades and Eaton fires in California killed 29 people, destroyed over 15,000 structures, and displaced more than 130,000. Simulations of California WUI communities by the UC Institute of Transportation Studies show that route and timing decisions dominate outcomes: phased evacuation cut travel

times by 13.5 to 70%, and reducing vehicle counts cut fire-exposure time by 27 to 60%. Operational fire spread models (Rothermel 1972; FARSITE, Finney 1998) produce the fire-arrival-time fields this problem uses.

1.2 Why it resists exact methods

Routing over a time-varying graph whose edges expire is NP-hard for multiple groups, and it must be re-solved as fire forecasts update. There is no gradient, and the objective has discontinuous failure points: one late segment turns a two-hour plan into a trapped group. A static shortest path is unsafe here, which our fire-blind baseline demonstrates by losing a group.

2 Background: Population-Based Optimization and ACO

A population-based metaheuristic keeps many candidate solutions that share information. The No Free Lunch theorem (Wolpert and Macready, 1997) says the algorithm must be justified by the structure of the problem. Here that structure is stepwise route construction on a graph with a per-step feasibility clock, which is how ACO already works.

Ant Colony Optimization (Dorigo, Maniezzo and Colorni, 1996) is based on real ant colonies. Ants lay pheromone as they walk; shorter paths are traversed more often, so they accumulate more pheromone and recruit more ants, while evaporation removes stale trails. Later variants include the Ant Colony System (Dorigo and Gambardella, 1997) and the MAX-MIN Ant System (Stützle and Hoos, 2000). The evaporation mechanism corresponds directly to roads that the fire closes.

2.1 The ACO equations

An ant at intersection i chooses the next feasible intersection j with probability

$$p_{ij} = \frac{\tau_{ij}^{\alpha}\,\eta_{ij}^{\beta}}{\sum_{k\in\mathcal{F}}\tau_{ik}^{\alpha}\,\eta_{ik}^{\beta}}, \qquad \eta_{ij} = \frac{1}{t_{ij} + \hat{t}_{\text{shelter}}(j)}, \tag{1}$$

where each term means:

- τ_{ij} (pheromone): the colony's memory, i.e. how much past successful ants favoured road (i, j).
- η_{ij} (heuristic): the ant's instinct, a quick score for how good (i, j) looks right now with no memory involved.
- α, β: how much weight the ant gives to memory (α) versus instinct (β) when the two disagree.
- $\mathcal{F}$: the feasible set, i.e. neighbours of i that can still be reached before the fire arrives.
- t_{ij}: the travel time along road (i, j).
- $\hat{t}_{\text{shelter}}(j)$: the estimated straight-line time from j onward to the nearest shelter.

η_{ij} is the reciprocal of $t_{ij} + \hat{t}_{\text{shelter}}(j)$, so roads that are quick to drive and point toward a shelter score higher. After each iteration, pheromone is updated by evaporation and then deposit:

$$\tau_{ij} \leftarrow (1-\rho)\,\tau_{ij} \quad \text{(evaporation)}, \tag{2}$$

$$\tau_{ij} \leftarrow \tau_{ij} + Q/L \quad \text{on the best plan's successful routes}, \tag{3}$$

where ρ is the evaporation rate, the fraction of pheromone lost each round so old trails fade and the colony can abandon roads the fire has cut off; Q is a fixed deposit constant; and L is the best plan's total penalized time, so a faster plan (smaller L) earns a larger Q/L reward and reinforces itself more strongly.

3 Related Work and Gap

ACO for fire evacuation already exists indoors. Xu, Huang, Liu, Li and Chen (*Journal of Building Engineering*, 2022) re-route building evacuees with an improved ACO driven by live fire-sensor data, and Wang (*Journal of Ambient Intelligence and Smart Environments*, 2022) combines bidirectional ACO with A* for high-rise fires. On outdoor road networks, Samadzadegan, Zarrinpanjeh and Schenk (Gi4DM 2011) applied ACO to post-earthquake routing with static damage costs. Outdoor wildfire evacuation with dynamically expiring roads exists without ACO: RESCUE (Tammali, Khanda, Satpathy, Shovan and Das, ICDCN 2026) solves a time-varying multi-objective shortest path with an edge-fire-risk function, and Zhang, Tan and Tiong (*Frontiers in Environmental Science*, 2025) combine SAR imagery, GNNs, Grey Wolf Optimization and multi-agent reinforcement learning for grassland-fire evacuation, reporting over 25% reduction in evacuation time. The specific combination implemented here, pure ACO over an outdoor road network whose segments expire as a wind-driven fire front arrives, did not appear in the literature review.

4 Solution Design

4.1 Fire model

The fire arrival time at a point $\mathbf{p}$ is its distance from the ignition $\mathbf{q}$ divided by a directional spread rate:

$$t_{\text{fire}}(\mathbf{p}) = \frac{\|\mathbf{p}-\mathbf{q}\|}{r(\mathbf{p})}, \qquad r(\mathbf{p}) = r_0\Big(1 + W\cos\theta(\mathbf{p})\Big), \tag{4}$$

where r_0 is the still-air spread rate, $W \in [0,1]$ is the wind strength, and θ is the angle between the spread direction and the wind. Downwind the fire moves up to $(1+W)\times$ faster and upwind it is slowed, which produces the wind-stretched ellipse of Huygens-type models (Richards 1990; FARSITE, Finney 1998), with Rothermel (1972) as the rate-of-spread ancestry: $R = R_0(1+\Phi_w+\Phi_s)$. The routing layer uses only this arrival-time field, so a higher-fidelity model such as the cellular automaton of Alexandridis et al. (2008) can replace it without changing the algorithm.

4.2 Encoding and feasibility

One ant proposes one full evacuation plan: for each group, an ordered list of intersections from its origin to a shelter. The feasibility rule, used by every method in the comparison, is

$$\text{cross}(u \to v \text{ at } t) \iff t \le t_{\text{fire}}(u) - m \ \wedge\ t + t_{uv} \le t_{\text{fire}}(v) - m, \tag{5}$$

with safety margin m. It is checked at every construction hop, so no infeasible plan is ever built, and an ant that reaches a dead end returns a failed route. Every generated scenario is checked for solvability by a fire-aware Dijkstra feasibility check before use; this check is used only as a scenario-solvability filter, not as a reported objective baseline.

4.3 Objective

Minimize the total penalized evacuation time,

$$T = \sum_{\text{groups}} \big(\text{arrival time at shelter}\big) + P \cdot \{\text{number of trapped groups}\}, \tag{6}$$

with $P = 8$ hours, so saving everyone outweighs speed without needing a second objective.

4.4 Parameters

The complete setup is best read as the rules of a board game: a map (the road network), a threat (the fire), players with a goal (the groups and shelters), a survival rule (the feasibility clock), and a solver (ACO). Table **??** lists every parameter grouped by role, with its value and its plain meaning.

Table 1: Simulation and algorithm parameters, grouped by role.

Parameter	**Value**	**Meaning**
Map (the road network)		
Grid size	11×11	Intersections (nodes) laid out as a grid.
Spacing	3.0 km	Nominal distance between neighbouring intersections.
Jitter	0.55	Random positional shift so the grid looks like an organic town, not a perfect lattice.
Roads removed	22%	Fraction of grid segments deleted at random, creating realistic dead ends and detours.
Driving speed	40 km/h	Constant vehicle speed used to turn road length into travel time.
Threat (the fire)		
Still-air spread	6 km/h	How fast the fire front moves outward with no wind.
Wind factor W	0.65	Wind strength; the front moves up to 65% faster downwind and slower upwind, stretching it into an ellipse.
Players and goal (evacuation)		
Groups	6	Neighbourhood groups (origins) that must reach safety.
Shelters	4	Designated safe zones each group can escape to.
Survival rule (feasibility)		
Safety margin m	0.12 h ($\approx$7 min)	A road is usable only if the group clears its far end at least this long before the fire arrives.
Fail penalty P	8 h	Hours added per trapped group, so saving everyone outweighs moving a few quickly.
Solver (ACO)		
Ants (N)	40	Candidate plans explored per iteration.
Iterations	120	Rounds of exploration before the final plan is chosen.
α	1.0	Weight on pheromone memory (trust in past successful ants).
β	3.0	Weight on the heuristic (instinct to head toward a shelter).
ρ	0.15	Evaporation rate; 15% of pheromone fades each round so stale trails are abandoned.
Q	10.0	Deposit constant; sets the reward size dropped on the best plan's roads.

Reproducibility. The wind factor $W = 0.65$ in Table **??** is fixed for every scenario; what is resampled per scenario (Section 6.2) is the wind *direction* $\theta \sim \text{Uniform}(0, 2\pi)$, together with the road network, ignition point, and origins. Concretely, given a scenario's random seed: intersection coordinates are jittered by independent $\mathcal{N}(0, \text{Jitter}^2)$ noise in each dimension, and each of the grid's candidate road segments is independently dropped with probability equal to the roads-removed rate (Bernoulli(0.22)); the ignition point is placed a fixed distance upwind of

the map centre along the freshly drawn θ and then jittered by independent $\mathcal{N}(0, (0.4\,\text{Spacing})^2)$ noise in each coordinate; the six origins and four shelters are each drawn uniformly without replacement, respectively, from the band of nodes at the 5th–25th percentile of fire-arrival time (origins, i.e. soon-to-be-threatened) and the band of nodes with the latest fire-arrival time (shelters, i.e. safest).

4.5 The overall loop

Putting the pieces above together, the solver repeats one loop for 120 iterations, alternating a local, per-ant construction phase with a global, per-iteration update phase:

1. **Local (each of the 40 ants, per group):** build a full route hop by hop, restricting each step to the feasible set $\mathcal{F}$ and sampling the next intersection with the transition probability (Eq. **??**); an ant with no feasible move left reports its group trapped.
2. **Score:** once every ant has finished every group, compute each plan's total penalized time T and keep the round's best plan.
3. **Global (once per iteration):** evaporate pheromone on all roads (Eq. **??**), then deposit the elitist reward Q/L only on the best plan's roads (Eq. **??**).

The incumbent best plan across all 120 iterations is returned as the final routing.

5 Results

5.1 Summary

In the demonstration scenario (seed 19; 121 intersections, 174 segments; six groups, four shelters) ACO evacuated all six groups in 5.020 hours and had effectively converged by iteration 43 of 120. Both baselines lost groups. Table **??** shows the comparison under the same feasibility rules. On the map the six routes visibly merge onto shared corridors, which is the pheromone consolidation shown directly.

Table 2: Total penalized evacuation time on the demonstration scenario (lower is better).

Method	Total time (h)	ACO improvement
ACO (this work)	5.020	—
Fire-aware greedy	18.956	73.5%
Fire-blind shortest path	11.715	57.1%

5.2 Statistical significance

Across 15 independent random scenarios (seeds 3000–3014; the road network, ignition point, wind direction, and group/shelter origins were independently resampled per scenario, while the wind factor $W = 0.65$ was held fixed, see Section 5.4), ACO gave a mean of 3.536 h (std 0.884) against fire-aware greedy's 12.176 h (std 13.699). The paired Wilcoxon signed-rank test (`scipy.stats.wilcoxon`) was run under its default configuration: a two-sided alternative, the `wilcox` zero-difference handling rule, and automatic method selection, which computes the exact null distribution for this sample size ($n = 15$) unless ties or zero differences among the paired scores force a fallback to the normal approximation, giving $p \approx 0.00043$, significant at 0.05; the test statistic and an effect-size measure were not retained from the run and are not reported here. The fixed trap penalty amplifies the variance of penalty-bearing failures, which is the tail

risk that matters most in an evacuation; the number of scenarios in which each method trapped a group was not separately retained, so no claim is made about a lower trapping frequency for ACO.

5.3 Figures

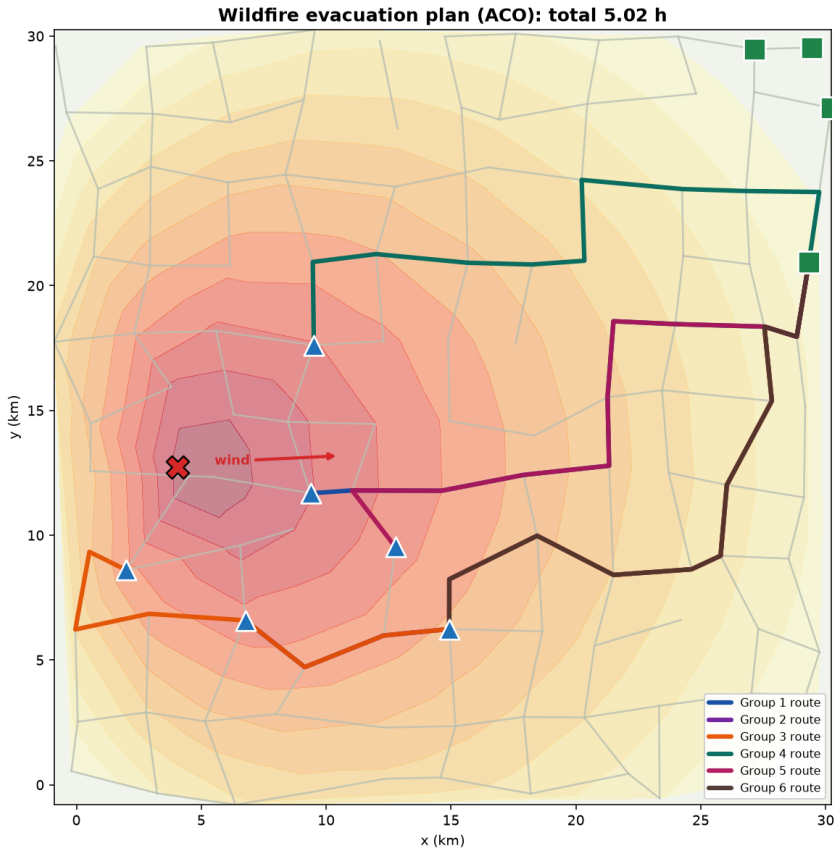


Figure 1: The evacuation plan. Warm contour bands show when the fire arrives everywhere (the wind-stretched ellipse), the red X is the ignition, blue triangles are origins, green squares are shelters, and the coloured routes are the ACO plan.

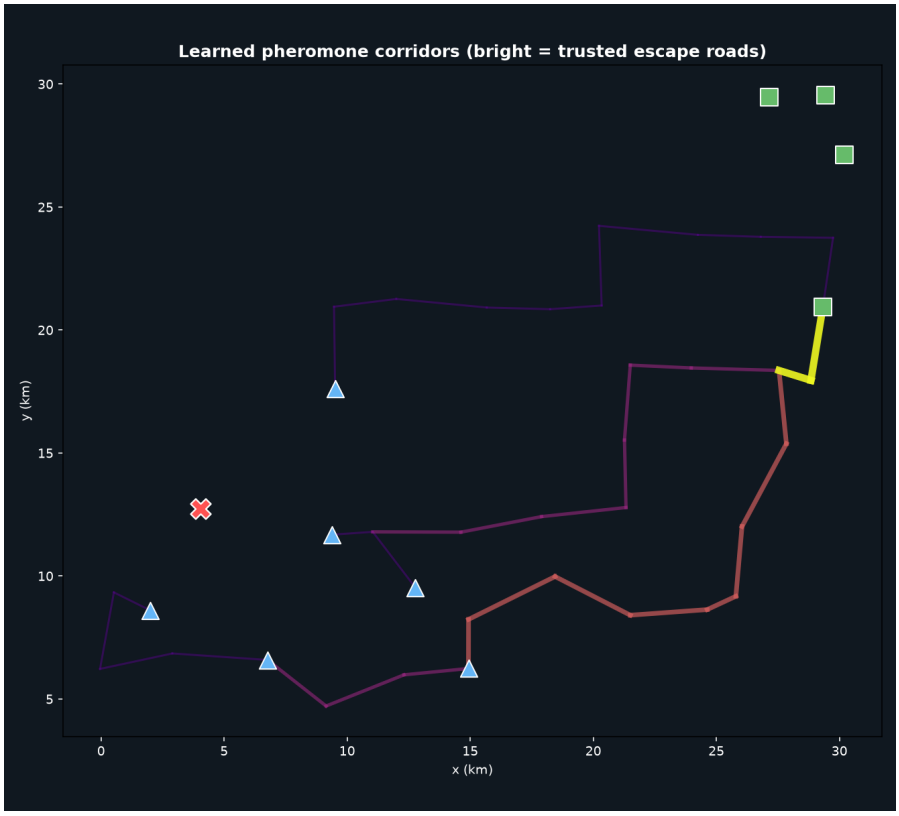


Figure 2: Learned pheromone corridors after convergence: bright edges are the roads the colony trusts to stay ahead of the fire.

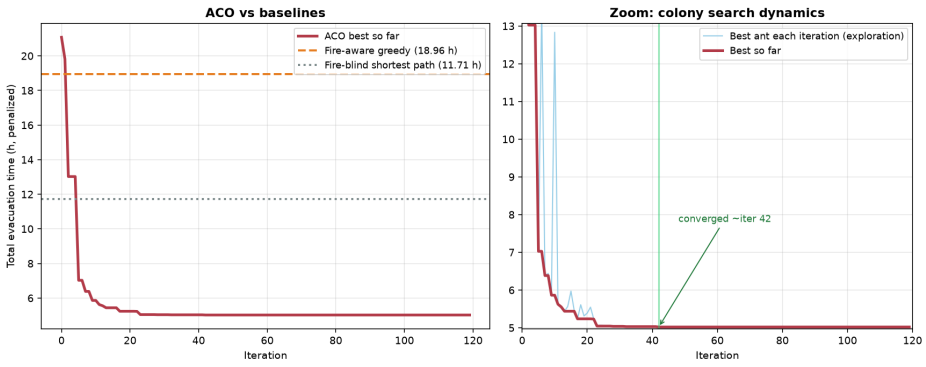


Figure 3: Convergence, in two panels. Left: ACO descends from 21 to 5 hours, crossing both baselines, which sit high after their trapped-group penalties. Right: a zoom on the colony's search dynamics; the light trace is the best ant each iteration (exploration) over the descending best-so-far staircase, converged near iteration 43.

The animation `evacuation_animation.gif` shows the fire growing with a charred core, a burning rim with glowing embers, and a bright front line. Road segments blacken as they burn, and six labelled cars drive their routes between house-marked origins and shelter-marked houses, turning green when safe, with a heads-up display counting minutes, groups safe, and roads lost.

6 Discussion and Conclusion

ACO fits this problem because an evacuation route is a stepwise graph walk against a clock, and checking feasibility during construction means every plan the search considers could actually be issued. All three methods are evaluated under the same crossing rule. However, the comparison also reflects differences in planning strategy and access to fire-arrival information, because the fire-blind baseline does not use the fire model during route selection: it plans a single shortest path with no fire information at all and is checked against the fire clock, hop by hop, only after that route has already been chosen, and none of the three methods replans once under way. Greedy uses the same feasibility rule and still gets trapped in dead-end pockets whose exits the fire closes, and the fire-blind baseline shows an ordinary shortest path losing a group outright. ACO's population spreads across corridors, and its pheromone concentrates on routes that are both fast and survive to the end, which cuts the mean time and, more importantly, the worst-case outcomes. Premature convergence is handled by evaporation, the stochastic transition rule, and elitist-but-eroding deposit.

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